



Krimiropa

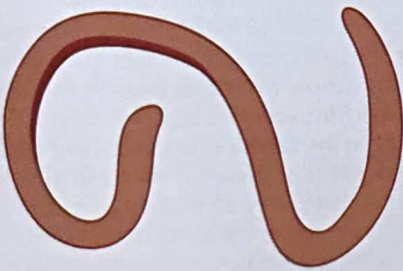
Learning Objectives

- ⇒ To understand the concept of Krimi
- ⇒ To understand the Nidana Panchaka of Krimi mentioned in Ayurveda literatures
- ⇒ To understand the relation between Nidana, Lakshana and Upadrava with fundamentals of physiology and pathology
- ⇒ To frame out the probable Samprapti Ghataka based on the progression of the disease

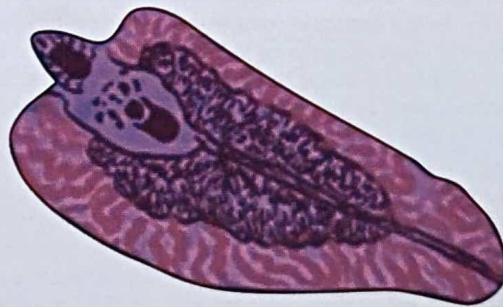
Krimi (Helminthes or worms) are very important among the infectious agents as they are responsible wide range of diseases in the living beings. They have large variation in size, complex life cycles and migratory habits. They

may be visible or invisible to the eyes. Depending upon the following derivation, *Krimi*, the one living being which moves around or which thrives on flesh and moves around.

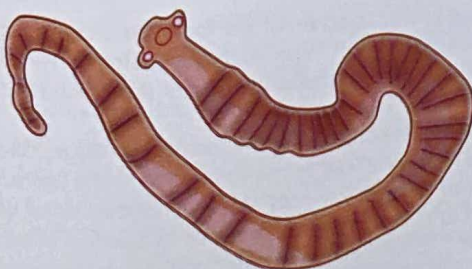
Types of Helminths (Parasitic Worms)



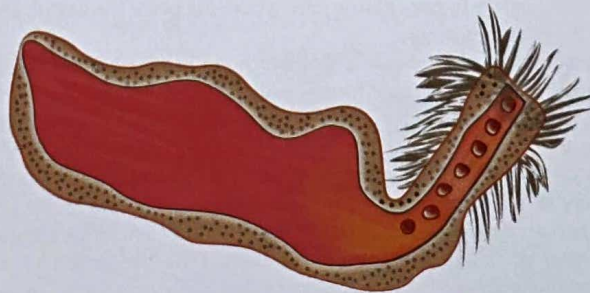
Roundworms: Can cause infection in the intestines or the body



Flukes: Generally infects the bile ducts, liver, or blood



Tapeworms: Infects the intestines



Thorny-headed worms: Mainly infects animals, rarely can infect humans

Figure 228: Types of Krimi

Nirukti

- ❖ *Krimi* is derived from क्रम् *Dhatu*
- ❖ क्रम् - पादविक्षेपः that means which moves with legs or limbs
- ❖ भ्रमे संप्रसारणेन च that which moves around (*Amarakosha*)
- ❖ क्रामति इति कृमिः (शब्दकल्पद्रुम) – the one which is moving

Definition of Krimi

- ❖ क्रव्ये मेध्याति क्रमते वा स्यात् सरणकर्मणः क्रामते वा॥ यक्षाचार्य means the one which thrives in *Kravya* (flesh) and performs movements is *Krimi*.

Samanya Nidana of Krimi

अजीर्णाध्यशनासात्म्यविरुद्धमलिनाशनैः।

अव्यायाम दिवास्वप्नगुर्वतिस्निग्धशीतलैः॥३॥

माषपिष्टान्नविदलबिसशालूकसेरुकैः।

पर्णशाकसुराशुतदधिक्षीरगुडेक्षुभिः॥४॥

पल्लानूपपिशितपिण्याकपृथुकादिभिः।

स्वाद्वम्लद्रवपानैश्च श्लेष्मा पित्तं च कुप्यति॥५॥

Su.Ut. – 54/3-5

अजीर्णभोजी मधुराम्लनित्यो द्रवप्रियः पिष्टगुडोपभोक्ता।

व्यायामवर्जी च दिवाशयानो विरुद्धभुक् संलभते

क्रिमींस्तु॥४॥

Ma.Ni. – 7/4

- ❖ **Ajirna¹** – when the individual is suffering from either *Amajirna* due to the vitiation of *Kapha Dosha* or *Vidagdhajirna* due to the vitiation of *Pitta Dosha*
- ❖ **Adhyashana²** – frequent consumption of food prior to the digestion of previously ingested meals
- ❖ **Asatmya Bhojana** – frequent consumption of unaccustomed food substances
- ❖ **Viruddha Bhojana** – frequent consumption of incompatible food substances
- ❖ **Malina Bhojana³** – frequent consumption of contaminated food substances
- ❖ **Avyayama** – not doing any of the physical activity
- ❖ **Divasvapna⁴** – sleeping during day time
- ❖ **Guru Bhojana⁵** – frequent consumption of substances which are heavy to digest
- ❖ **Snigdha Bhojana⁶** – frequent consumption of substances which are unctuous in nature
- ❖ **Shitala Bhojana⁷** – frequent consumption of substances which are cold in nature

G
K

- 1 When an individual is having *Ajirna* (indigested state), that will lead to proliferation of the microorganisms in the stomach. These microorganisms may enter the stomach through either poor oral hygiene or through contaminated food. Most of the individuals in India are having the carbohydrate rich diet which is considered to be major source of multiplication of microbes in the stomach. This may increase the infection in the stomach. As there is increased time period for the closure of pyloric antrum, these microorganisms proliferated in the stomach may cause gastric mucosal damage, so that macromolecules and intermediate molecules retained in the stomach because of poor digestive juice may enter the interstitial compartment of alimentary canal, which further enters into circulatory system through lymphatic system.
- 2 The pathological events occurring in this type of exposure is similar to the mechanism described in the context of *Ajirna*.
- 3 When an individual is having the excess exposure to these kinds of aetiological factors, there will be inflammation of the mucosa of the stomach takes place as these substances may be stored in the stomach at least 2 hours or so. This may further leads to the inflammation of the small intestine as they start to move into lower alimentary canal. These inflammatory reactions will release inflammatory markers which are nothing but proteins, which attract more amount of fluid portion from capillaries, which increases the capacity of the stomach, and as there is more alkalinity of the gastric juice, the substances are not getting digested due to the persistent inactivation of pepsinogen into pepsin. Due to the persistent indigestion of carbohydrate rich food, in the stomach, there will be proliferation of microorganisms leading to infection. But infection occurs in the stomach more frequently as the food substance may have more duration of contact to the inflamed tissue, which further leads to the transport of inflammatory markers into the blood via absorption in the stomach.
- 4 These activities will be responsible for poor gastro intestinal motility and this will lead to the proliferation of microorganisms in the alimentary canal. This is responsible for the damage to the gastro intestinal mucosa.
As per fundamentals of Ayurveda, there is increase of *Guru* quality in the body and there is decrease of *Chala* quality in the body. Due to these activities, the conversion of macromolecules to intermediate and micro molecules are not happening in the stomach and small intestine secondary to poor gastric juice where the pH of the gastric juice will become alkaline. Because of the above said reasons, proliferation of the microbes occurs.
- 5 Consumption of food substances which are heavy to digest is responsible for delay in the digestion of the food and there is poor gastro intestinal motility which is further leading to the proliferation of microorganisms in the alimentary canal leading to gastro intestinal mucosal damage. From the point of western physiology point of view, conversion of macro molecules into intermediate or micromolecules will not occur.
As per Ayurveda fundamentals, the excess consumption of *Guru* quality substances is responsible for the increase of *Guru* quality in the body. When we think of the analysis of *Dosha*, exposure to this kind of aetiological factor will lead to the vitiation of *Kapha Dosha*.
- 6 When an individual consumes more amount of unctuous substances, that will lead to the heaviness of the alimentary canal which will further leading to poor intestinal motility. Thus, the person will have proliferation of microorganisms develops as a sequel of poor gastro intestinal motility. In this occasion, reason is attributable to the non conversion of macro molecules into intermediate and micro molecules. As per fundamentals of Ayurveda, increased consumption of *Snigdha* substances will lead to the increase of *Snigdha* quality in the body, thus there will be abnormal increase of *Kapha Dosha* in the body.
- 7 When an individual consumes *Shita* substances, that will lead to the poor blood circulation to the alimentary canal, thereby there is decreased digestive juices in the stomach and small intestine. This leads to the proliferation of microorganisms in the stomach and small intestine as there is poor secretion of hydrochloric acid which is fundamental to any of the anti infectious activities in the alimentary canal

- **Masha, Pishtanna, Vidala, Bisa, Shaluka Seruka⁸** – substances which are prepared from blackgram, corn flour or any kind of flour, pulses, *Bisa, Shaluka* and *Kaseruka*
- **Parna Shaka⁹** – leafy vegetables
- **Sura, Shukta, Dadhi, Kshira, Guda and Ikshu¹⁰** – consumption of intoxicating substances such as *Sura, Shukta*, curds, milk, jaggery and sugarcane and sugarcane products
- **Palala¹¹** – consumption of paste of sesamum seeds
- **Anupa Pishita¹²** – consumption of meat of animals of marshy lands
- **Pinyaka and Prithuka** – excess consumption of these substances
- **Svadu and Amla Rasa Sevana¹³** – consumption of substances having sweet and sour taste
- **Drava Pana¹⁴** – excess consumption of liquid substances

Krimi as a cause of contagious disease¹⁵

प्रसङ्गाद्वात्रसंस्पर्शाग्निःश्वासात्सहभोजनात्।

सहशय्यासनाच्चापि वस्त्रमाल्यानुलेपनात्॥3 3॥

Su.Ni. – 5/33

- **Prasanga** – sexual contact
- **Gatra Samsparsa** – touching the contaminated parts of the body
- **Nishvasa** – inhaling the contaminated air or breath of infected individual
- **Sahabhajana** – consuming the food together
- **Saha Shayya** – sharing the bed with infected individual
- **Saha Asana** – sitting together
- **Vastra Malanulepana** – sharing the clothes, garlands and ornaments of infected individual

Vishesha Nidana of Krimi

माषपिष्टात्सल्लवणगुडशकैः पुरीषजाः।

मांसमत्स्यगुडक्षीरदधिशुक्तैः कफोद्भवाः॥5 1॥

and the main content of most of the individual is carbohydrate rich diet. microorganisms proliferate very easily and rapidly in carbohydrate rich diet.

As per fundamentals of Ayurveda, there is increase of Shita quality occurs as a result of excess consumption of Shita substances by an individual. This will lead to the abnormal increase of Kapha Dosha in the body.

All the above said substances are having Guru, Snigdha, Shita qualities and Madhura Vipaka. As per the fundamentals of Ayurveda, the person will have abnormal increase of Kapha Dosha. As per the principles of western physiology, because of the consumption of the substances having above said qualities, there will be poor peristaltic movement in the alimentary canal which in turn is responsible for the proliferation of microorganisms in the alimentary canal. This event in turn is responsible for the damage of the mucosa of the alimentary canal. For example, oral mucosa can be damaged if proper oral hygiene after food intake is not followed appropriately. Most of the Indians are having carbohydrate rich diet. Rapid multiplication of microorganisms is likely to occur secondary to poor oral hygiene or excess carbohydrate diet intake. Thus, mucosal damage can occur in any part of the alimentary canal.

In this occasion, the term leafy vegetables consumption can be attributable to presence of worms in the leaf of vegetables or spoiled vegetables. Thus, improper washing of leafy vegetables may be the probable reason for the exposure of Krimi in susceptible individuals.

Consumption of these substances may lead to depletion of accumulated fat from different parts of the body as the person is consuming less carbohydrate rich substances. Initially, this will lead to the causation of inflammation, as these substances causing irritation to the gastrointestinal mucosa. When an individual consumes macronutrient diet, then that will lead to the occurrence of infection because of poor dental hygiene and consumption of macronutrients, glucose in particular. Thus, entry of microorganisms in this occasion will lead to the proliferation of the microorganisms with less resistance from the body due to reduced immunity. These substances which are causing intoxication are usually responsible for the inflammation of the alimentary canal as these substances are having Amla Rasa, Laghu Guna and Amla Vipaka. When inflammation starts in the alimentary canal, that will lead to the damage of the mucosa of the alimentary canal.

Excess consumption of sesamum seeds leads to the vitiation of Pitta Dosha in the body. Excess vitiation of Pitta and Kapha Dosha will lead to the manifestation of Krimi in the healthy or diseased individual. This is attributable to the inflammation of the alimentary canal which further leads to the damage of the gastrointestinal mucosa.

When an individual consumes meat of animals of marshy animals will lead to the increase of Kapha Dosha in the body. Further, there are other reasons for the multiplication of microorganisms, i.e. improperly cooked meat by an individual.

Excess consumption of Madhura and Amla Rasa substances will lead to the increase of Kapha Dosha in the body. The consumption of sweet substances will be the breeding diet for the proliferation of microorganisms and consumption of sour substances will lead to the inflammation of the alimentary canal, thereby results in the poor immunity of the individual.

Excess consumption of Drava (liquid) substances will lead to the increase of Drava quality in the body. This will lead to the dilution of gastric juice (dilution of hydrochloric acid) which in turn is responsible for inactivation of pepsinogen. It is being said that the optimum range of pH of gastric juice for the activation of pepsinogen into pepsin is 1.9 to 3.8; pH beyond 3.8 leading to the movement of gastric juice towards alkalinity, will lead to the multiplication of microbes in the alimentary canal. Thus, excess liquid consumption will lead to the proliferation of microorganisms in the stomach leading to the progression of infective pathology.

These are the different routes of entry of microorganisms into the body. As per the description of Sushruta, urethra, vagina, skin contact, expired air, certain habits where there is increased risk of contagious disease manifestation such as consuming food together (through alimentary canal), sleeping and sitting together (through skin).

As per current understanding, all natural orifices present in the body such as mouth, nostrils, eyes, ears, anus, urethra, and in females, vagina and nipples are sources of entry of microorganisms if proper care of these orifices are not being done. If there is any cut in the skin of the infected individual, may lead to proliferation of infection and if such individual is having contact with the person who is having cut in the skin may lead to spread of infection.

विरुद्धाजीर्णशाकाद्यैः शोणितोत्था भवन्ति हि।

Ma.Ni. – 7/5,6

Purishaja Krimi are produced by excess consumption of blackgram items, flour items, sour and salty substances, substances prepared by jaggery.

Kaphaja Krimi are proliferated by excess consumption of flesh, fishes, jaggery, products prepared by milk and curds and *Shukta*, a intoxicating substance.

Raktaja Krimi are proliferated by excess consumption of non homologous substances, consumption of food substances in case of state of indigestion.

Samprapti of Krimi

स्वाद्वल्लद्रेवपानैश्च श्लेष्मा पित्तं च कुप्यति॥5॥

कृमीन् बहुविधाकारान् करोति विविधाश्रयान्।

आमपकाशये तेषां प्रसवः प्रायशः स्मृतः॥6॥

Su.Ut. – 54/5-6

- ⊗ There will be the vitiation of *Kapha* and *Pitta*
- ⊗ This is leading to genesis of various kinds of *Krimi* which are having different forms or shapes, localizing in different sites.
- ⊗ *Amashaya* is the site of *Shleshmaja Krimi*, *Pakvashaya* is the seat of *Purishaja Krimi* and *Dhamani* is the seat of *Raktaja Krimi*.

Samprapti Ghataka of Krimi

Dosha : *Kapha* and *Pitta* Dosha

Dushya : *Rasa*, *Rakta*, *Purisha*

Srotas : *Rasavaha*, *Raktavaha*, *Purishavaha* Srotas

Srotodushti Prakara: *Sanga*, *Vimargagamana*

Agni : *Jatharagnimandya* followed by *Rasa* and *Rakta* *Dhatvagnimandya*

Ama : *Jatharagnimandya* *janya Ama*

Udbhava Sthana : *Amashaya* (*Shleshmaja* and *Raktaja Krimi*) and *Pakvashaya* (*Purishaja Krimi*)

Sanchara Sthana : *Sarvasharira*

Vyakta Sthana : *Tvak*, *Guda*, *Vaktra*

Rogamarga : *Abhyantara* and *Bahya* *Rogamarga*

Sadhyasadyata : *Keshada* and *Romada* varieties of *Krimi* are incurable and rest of the *Krimi* are curable.

Types of Krimi

- ⊗ *Sankhya Samprapti* of *Krimi* as per *Charaka*, *Sushruta* and *Madhava Nidana* is 20. There is difference of opinion among sub types.

2 types:

- **Vaikarika** – pathogenic
- **Sahaja** – non pathogenic

2 types of Vaikarika Krimi:

- **Bahya** – 2
- **Abhyantara** – 18

20 types:

- **Bahya** – 2: *Yuka* and *Pipilika*
- **Shonitaja** – 6: take shelter in *Raktavahi Dhamani*; *Keshada*, *Lomada*, *Lomadvoipa*, *Saurasa*, *Audumbara*, *Jantumatara*
- **Shleshmaja** – 7: take shelter in *Amashaya*; *Antrada*, *Udarada*, *Hridayachara*, *Churu*, *Darbhapushpa*, *Saugandhika*, *Mahaguda*
- **Purishaja** – 5: take shelter in *Pakvashaya*; *Kakeruka*, *Makeruka*, *Leliha*, *Sashula*, *Sausurada*
- ⊗ As per *Sushruta*, 20 types of *Krimi* are subdivided into 3 categories.
- **Purishaja Krimi** – 7: *Ajava*, *Viyava*, *Kipyra*, *Chipya*, *Gandupada*, *Chooru*, *Dvimukha*
- **Kaphaja Krimi** – 5: *Darbhapushpa*, *Mahapushpa*, *Praluna*, *Chipita*, *Pipilika*, *Daruna*
- **Shonitaja Krimi** – 7: *Keshada*, *Romada*, *Nakhada*, *Dantada*, *Kikkisa*, *Kushthaja*, *Parisarpa*

Samanya Lakshana of Abhyantara Krimi

(ज्वरो विवर्णता शूलं हृद्रोगः सदनं भ्रमः॥18॥

भक्तद्वेषोऽतिसारश्च संजातकृमिलक्षणम्।)

Su.Ut. – 54/18-19; Ma.Ni. – 7/6,7

- ⊗ **Jvara**¹⁶ : Raised body temperature
- ⊗ **Vivarnata**¹⁷ : Discolouration of the body

- G K**
- 16 As stated earlier in the textbook, any conditions which lead to either increased heat production or reduced heat loss from the body will lead to the raise of body temperature. Depending upon the exposure of microorganism into the body through any route will lead to the proliferation of microbes and later that will lead to the increase of WBCs in the body. This event is responsible for the action of WBCs on microbes and this will be responsible for the raised body temperature. This situation will alter the Sterling forces, leading to sludging of blood and thus the affected region will have swelling of the region.
- Any conditions which lead to septicemia or septic shock (this can be attributable to *Shonitaja Krimi*) such as
- Peritonitis caused by spread of infection from uterus and uterine tubes and sometimes caused by instrumental abortion under unsterile condition
 - Peritonitis caused by rupture of gastro intestinal tract or caused by intestinal disease or caused by wounds
 - generalised bodily infection resulting from spread of skin infection such as streptococcal or staphylococcal infection
 - generalised gangrenous infection resulting from gas gangrene bacilli, spreading first through peripheral tissues and finally by way of the blood to the internal organs, liver in particular
 - Infection spreading into the blood from the kidney or urinary tract, often caused by colon bacilli
- 17 when an individual is having uncontrolled infection which can enter through any of the natural orifices or through trauma, following mechanisms are observed in the individuals.
1. High grade of fever

- ❖ **Shula**¹⁸ : Pricking type of pain
- ❖ **Hridroga**¹⁹ : Heart disease
- ❖ **Sadana**²⁰ : Sluggish movement of the body
- ❖ **Bhrama**²¹ : Giddiness
- ❖ **Bhaktadvesha**²² : Aversion to consume food substances
- ❖ **Atisara**²³ : Diarrhoea

Bahya Krimi

तत्र मलो बाह्यश्चाभ्यन्तरश्च।

तत्र बाह्यमलजा- तान मलजान् संचक्ष्महे।

तेषां समुत्थानं सृजाव- र्जनं स्थानं - केशश्मश्रुलोमपक्ष्मवासांसि
संस्था- नम् - अणवस्तिलाकृतयो बहुपादाश्चः वर्णाः कृष्णाः,
शुक्लाश्च; नामानियुक्ताः, पिपीलिकाश्च प्रभावः-2 कण्डूजननं,
कोठपिडकाभिनिर्वर्तनं च; चिकित्सितं तु खल्वेषामपकर्षणं,

मलोपघातः, मलकराणां च भावानामनुपसेवन मिति॥10॥

Ca.Vi. - 7/10

तिलप्रमाणसंस्थानवर्णाः केशाम्बराश्रयाः॥12॥

बहुपादाश्च सूक्ष्माश्च युक्ता लिक्षाश्च नामतः।

Ma.Ni. - 7/2-3

द्विधा ते कोठपिडकाकण्डूगण्डान् प्रकुर्वते॥13॥

Ma.Ni. - 7/3

अजीर्णभोजी मधुराम्लनित्यो द्रवप्रियः पिष्टगुडोपभोक्ता।

व्यायामवर्जी च दिवाशयानो विरुद्धभुक् संलभते किर्मिस्तु॥14॥

Ma.Ni. - 7/4

- ❖ *Bahya Krimi* are *Tila* (sesamum) seed sized and similar in shape and colour, which are harboured in hairs and clothes, have multiple limbs, and microscopic or minute in appearance. They lead to the urticarial maculo-papular rashes, pustules, itching and hard lesion.

- ❖ These *Krimi* are manifested due to the excess consumption in the stage of *Ajirna* (indigestion),

2. Marked vasodilatation throughout the body, especially in the infected tissues.

3. High cardiac output in perhaps half of the patients, caused by arteriolar dilatation in the infected tissues and by high metabolic rate and vasodilatation elsewhere in the body, resulting from bacterial toxin stimulation of cellular metabolism and from high body temperature.

4. Sludging of the blood, caused by red cell agglutination in response to degenerating tissues

5. Development of micro clots in widespread areas of the body, a condition called disseminated intravascular coagulation. This will lead to the usage of blood clotting factors to be used up, so that bleeding occurs in many tissues, especially in the gut wall of the intestinal tract.

Here Vivarnata refers to the discolouration of the skin in the form of redness of the region or yellowish as a result of accumulation of pus and subcutaneous tissues which manifest as a result of development of either purpuric patches etc.

18 The term *Shula* refers to pricking type of pain. When an individual is having intestinal worms, he will have pricking type of pain in the abdominal region. If he has infection in the skin, he will have inflammation in the skin which may be presented with pricking type of pain in the skin. As these worms may accumulate in the small or large intestine, this may lead to partial or complete intestinal obstruction, which leads to colicky type of pain in the abdominal region.

19 This can be attributable to cardiogenic shock or hypovolemic shock. This may be manifested due to shift of the more amount of the fluid to the afflicted part.

20 The term *Sadana* refers to sluggish movements of the body parts. When an infection occurs inside the body, that will lead to the poor blood flow in the small blood vessels in the circulatory system. This is called sludging of blood. In this situation, there will be continuation of tissue metabolism despite poor blood flow leading to large amount of acids – carbonic acid and lactic acid to accumulate locally enhancing the local acidity of the tissues. This acid along with other degeneration products from the ischemic tissues causes local blood agglutination. Thus, the person will have poor blood movement along the body, leading to the sluggish movement of the body parts.

The probable causes of septic shock are

- Cardiac depression or poor contractility of myocardial muscles by release of endotoxin
- Vasomotor failure
- Slugged blood – poor blood flow
- Increased capillary permeability
- Release of toxins by ischemic tissue
- Generalised cellular deterioration
- Development of acidosis either lactic acidosis or excess concentration of carbonic acid in the body

21 This is attributable to the development of hypovolemic shock. This may occur as a result of poor cerebral blood flow which results in the manifestation of syncopal attacks. This is manifested due to the sludging of blood (poor blood flow in the circulatory system). Probable causes of this type of shock (hypovolemia induced) are

- Excessive sweating
- Fluid loss in severe diarrhoea or vomiting
- Excess loss of fluid by nephrotic kidneys
- Inadequate intake of fluids and electrolytes
- Destruction of adrenal cortex with loss of aldosterone secretion

22 This can be attributable to dilution of gastrointestinal secretion. This further leads to the proliferation of microorganisms in the alimentary canal. This further leads to the occurrence of gastric mucosal damage leading to the poor gastrointestinal secretion may lead to the aversion to consume food substances.

23 This can be attributable to the proliferation of microorganisms in the lower alimentary tract. This will lead to the inflammation of the small and large intestines which further leads to the excess secretion of the small and large intestinal secretions which is responsible for increase of bulk of faecal matter. As there is inflammation in large intestine, there will be increased peristaltic movements are seen, this will lead to the occurrence of diarrhoea.

excess consumption of *Madhura* (sweet), *Amla* (sour) *Rasa* substances, excess consumption of *Drava* (liquid) substances, excess consumption of substances prepared from flour and jaggery, the people who are averse to do physical activities and sleeping in day time and the people who has the habit of incompatible substances.

Kaphaja Krimi

श्लेष्मजाः क्षीरगुडतिलमत्स्यानूपमांसपिष्टान्नप- रमान्न-
कुसुम्भस्नेहाजीर्णपूर्ति क्लिन्नसंकीर्णविरुद्धासात्यं भोजन समुत्थानाः
तेषामामाशयः स्थानं, प्रवर्धमानास्तूर्ध्वमधो वा विसर्पन्त्युभयतो
वाः संस्थान वर्णविशेषास्तु- श्वेताः पृथुब्रध्नसंस्थानाः केचित्
केचिद्वृत्तपरिणाहा गण्डूपदाकृतयः श्वेतास्तानावभासाश्च केचिदणवो
दीर्घास्तन्वाकृतयः श्वेताः तेषां त्रिविधानां श्लेष्मनिमित्तानां क्रिमीणां
नामानि - अन्नादाः, उदरादाः, हृदयंचराः, चुरवः, दर्भपुष्पाः,
सौगन्धिकाः, महागुदाश्चेति प्रभावो- हल्लासः, आस्यसंस्त्रवणम्,
अरोचकाविपाकौ, ज्वरः, मूर्च्छा, जृम्भा, क्षवथुः, आनाहः, अङ्गमर्दः,
छर्दिः का, पारुष्यं चेति॥ 2 ॥

Ca.Vi. - 7/12

Nidana of Kaphaja Krimi

मांसमत्स्यगुडक्षीरदधिशुक्तैः कफोद्धवाः॥ 5 ॥

Ma.Ni. - 7/5

Consumption meat, fish, jaggery, milk and milk products, curds, alcoholic beverages in excess are mentioned as the causative factors for the growth of *Kaphaja Krimi* in the body.

Lakshana of Kaphaja Krimi

हल्लासमास्यस्त्रवणमविपाकमरोचकम्॥ 10 ॥

मूर्च्छाच्छर्दिज्वरानाहकाक्षवथुपीनसान्। Ma.Ni. - 7/10,11

- ⊗ These sets of *Krimi* are proliferated in *Amashaya* due to the vitiation of *Kapha Dosha* and move throughout the body.
- ⊗ These *Krimi* are either flat or elongated. A few *Krimi* are like earthworm, elevated, curved and some are short.
- ⊗ They are white when they have not consumed anything and later transform into coppery colour after the consumption of sucking *Rasa* and *Rakta* from the gut.
- ⊗ They are seven in number.
- ⊗ They are responsible for the manifestation of nausea, excess salivation, indigestion, tastelessness, altered state of consciousness, vomiting, fever, distension of abdomen, emaciation, sneezing, running nose.

Raktaja Krimi

शोणितजानां तु खलु कुष्ठैः समानं समुत्थानं; स्थानं -
रक्तवाहिन्यो धमन्यः; संस्थानम् -

अणवो वृत्ताश्चापादाश्च, सूक्ष्मत्वाच्चैके भवन्त्यदृश्याः वर्णः-
ताम्रः नामानि - केशादा, लोमादा, लोमद्वीपाः, सौ-
रसा, औडुम्बरा, जन्तुमातरश्चेति; प्रभावः -
केशश्म श्रुनखलोमपक्ष्मापध्वंसः, व्रणगतानां च हर्षकण्डूतोदसंसर्पणानि,
अतिवृद्धानां च त्वक्सिरास्नायुमां- सतरुणास्थिभक्षणमिति;
चिकित्सितमप्येषां कुष्ठैः समानं, तदुत्तरकालमुपदेक्ष्यामः॥ 1 ॥

Ca.Vi. - 7/11

केशरोमनखादाश्च दन्तादाः किक्किशास्तथा।

कुष्ठजाश्च परीसर्पा ज्ञेयाः शोणितसंभवाः॥ 5 ॥

Su.Ut. - 53/15

ते सरक्ताश्च कृष्णाश्च स्निग्धाश्च पृथ्वस्तथा।

Su.Ut. - 53/16

रक्ताधिष्ठानजान् प्रायो चिकारान् जनयन्ति ते॥ 6 ॥

Su.Ut. - 53/16

Nidana of Shonitaja Krimi

विरुद्धाजीर्णशाकाद्यैः शोणितोत्था भवन्ति हि।

Su.Ut. 53/18; Ma.Ni. 7/6

Excess consumption of *Viruddha Ahara* (18 types of *Viruddha* as per *Charaka*), *Ajirna* (a state of indigestion) and certain types of vegetables are considered as the causative factors for the proliferation of *Shonitaja Krimi* in the body.

Lakshana of Shonitaja Krimi

षट् ते कुष्ठैककर्माणः

Ma.Ni. - 7/12

- ⊗ *Shonitaja Krimi* are present in *Raktavahi Dhamani* and circulate along with *Rakta* (blood).
- ⊗ These are microscopic in appearance, circular without limbs and some of the *Krimi* are invisible and colour of these *Krimi* is coppery colour. Some of the *Krimi* are having blackish discolouration.
- ⊗ As per *Charaka*, the clinical features of *Raktaja Krimi* are similar to *Kushtha* (group of skin diseases). The clinical features observed in case of affliction of *Raktaja* or *Shonitaja Krimi* are destruction of scalp hairs, abnormal sensation of pricking type of pain, horripulation, itching, in advanced stages, there will be destruction of skin, blood vessels, ligaments, muscles and tendons.

Purishaja Krimi

पुरीषजास्तुल्यसमुत्थानाः श्लेष्मजैः; तेषां स्थानं पक्काशयः, ते
प्रवर्धमानास्तूर्ध्वमधो विसर्पन्ति, यस्य पुनरामाशयाभिमुखाः स्युर्यदन्तरं
तदन्तरं तस्यो- द्वारनिःश्वासाः पुरीषगन्धिनः स्युः; संस्थानवर्ण-
विशेषास्तु- सूक्ष्मवृत्तपरीणाहाः श्वेता दीर्घा ऊर्णाशु- संकाशाः केचित्
केचित् पुनः स्थूलवृत्तपरीणाहाः श्यावनीलहरितपीताः; तेषां नामानि
ककेरुकाः, मकेरुकाः, लेलिहाः; सशूलकाः, सौसुरादाश्चेति; प्रभावः
--- पुरीषभेदः, कार्यं, पारुष्यं, लोमहर्षाभि- निर्वर्तनं च त एव चास्य
गुदमुखं परितुदन्तः कण्डू चोपजनयन्तो गुदमुखं पर्यासते, त एव
जातहर्षा गुदनिष्क्रमणमतिवेलं कुर्वन्ति इत्येष श्लेष्मजानां पुरीषजानां
च किमीणां समुत्थानादिविशेषः॥ 3 ॥

Ca.Vi. - 7/13

Nidana of Purishaja Krimi

माषपिष्टाम्ललवणगुडशकैः पुरीषजाः।

Excess consumption of black gram items, flour substances, sour and salty substances, jaggery and certain vegetables are considered as the causative factors for the multiplication of *Purishaja Krimi*.

Ma.Ni. – 7/5

Lakshana of Purishaja Krimi

विद्भेदशूल विष्टम्भकार्श्यपारुष्यपाण्डुताः।

रोमहर्षाग्निसदनं गुदकण्डूविमार्गगाः॥16॥

Ma.Ni. – 7/16

- *Purishaja Krimi* are arising from *Pakvashaya*, usually moving downwards. When increased in number, they move towards *Amashaya*, then produce smell of faeces in the mouth, belchings and expiration
- They are thick, round, blue, yellowish, white or black in colour.

- *Purishaja Krimi* are five in number.
- The clinical features of *Purishaja Krimi* are diarrhoea, pain abdomen, poor peristaltic movements, emaciation, roughness of the skin, yellowish white discoloration, horripilation, reduced appetite, itching around anal region.

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Sadhyasadyata of Krimi

दृश्यास्त्रयोदशाद्यास्तु कृमीणां परिकीर्तिताः॥19॥

केशादाद्यास्त्वदृश्यास्ते द्वावाद्यौ परिवर्जयेत्।

Su.Ut. – 54/19,20

- Out of 20 types of *Krimi*, 13 varieties of *Purishaja* and *Kaphaja Krimi* are visible and 7 varieties of *Krimi* are invisible. Among these *Krimi*, *Keshada* and *Romada*²⁴ varieties of *Krimi* are very difficult to cure and other varieties of *Krimi* are treatable.

Key summary points of the chapter

- क्रव्ये मेध्याति क्रमते वा स्यात् सरणकर्मणः क्रामते वा॥ यक्षाचार्य means the one which thrives in *Kravya* (flesh) and performs movements is *Krimi*.
- The causative factors which vitiate the *Kapha* and *Pitta* Dosha are responsible for the manifestation of *Krimi* in the body.
- **Sankhya Samprapti of Krimi is 20.** They are further subclassified into 4 –
 - *Bahya* (2),
 - *Shleshmaja* (7),
 - *Shonitaja* (6) and
 - *Purishaja* (5) *Krimi*.
- **2 Bahya Krimi** are *Yuka* and *Pipilika*.
- **7 Shleshmaja Krimi** are *Antrada*, *Udarada*, *Hridayachara*, *Churu*, *Darbhapushpa*, *Saugandhika*, *Mahaguda*.
- **6 Shonitaja Krimi** are *Keshada*, *Lomada*, *Lomadvipa*, *Saurasa*, *Audumbara* and *Jantumata*.
- **5 Purishaja Krimi** are *Kakeruka*, *Makeruka*, *Leliha*, *Sashula* and *Sausurada*.
- **Sushruta has also mentioned 20 types of Krimi.**
- As far as *Vidhi Samprapti* of *Krimi* is concerned, 2 types of *Krimi* *Sahaja* and *Vaikarika*.
 - *Vaikarika Krimi* needs special attention as these are responsible for various pathological events in the body.
 - 2 types of *Vaikarika Krimi* are *Bahya* and *Abhyantara*.
 - *Bahya* types of *Krimi* can be removed by external means such as *Yantra* etc whereas *Abhyantara Krimi* require internal medicines to eliminate these *Krimi* in the body.
- (ज्वरो विवर्णता शूलं हृद्रोगः सदनं भ्रमः॥18॥
भक्तद्वेषोऽतिसारश्च संजातकृमिलक्षणम्।) Su.Ut. – 54/18-19; Ma.Ni. – 7/6,7 *Jvara*, *Vaivarnya*, *Shula*, *Hridroga*, *Sadana*, *Bhrama*, *Bhaktadvesha* and *Atisara* are the *Lakshana* observed when an individual is having *Krimi* inside the body.
- Among the *Krimi*, *Keshada* and *Romada* varieties of *Shonitaja Krimi* are very difficult to cure whereas the other types of *Krimi* are comparatively easier to cure.

²⁴ *Keshada* and *Romada* are the varieties of *Shonitaja Krimi*. Here, we can attribute that there is a possibility of septicemia or septic shock. This type of shock is considered to be the most frequent cause of shock related death in the hospitals. Refer the context of septic shock explained in this chapter. In the initial stages of septic shock, the patient usually does not have signs of circulatory collapse but only signs of bacterial infection. As the infection becomes more severe, the circulatory system usually becomes involved either because of direct extension of the infection or secondarily as a result of toxins from the bacteria, with resultant loss of plasma into the infected tissues through deteriorating blood capillary walls.

Self-Assessment: Questions for Reviews

Long essay questions

- Define Krimi. Explain the Samanya Nidana, Samprapti and Samanya Lakshana and types of Krimi and Sadhyasadyata of Krimi.

Short essay questions

- Explain the Vidhi Samprapti of Krimi
- Explain Shonitaja (Raktaja) Krimi
- Explain Shleshmaja (Kaphaja) Krimi
- Explain Purishaja Krimi

Short answers

- Write the definition of Krimi
- Enumerate 20 types of Krimi
- Write a short note on Shonitaja Krimi
- Write a short note on Kaphaja Krimi
- Write a short note on Purishaja Krimi

MCQs

- Sankhya Samprapti of Krimi as per Charaka is
 - 2;
 - 20;
 - 7;
 - 10

Answer: (b)

- The number of Purishaja Krimi as per Charaka is
 - 2;
 - 7;
 - 6;
 - 5

Answer: (d)

- Darbhapushpa is the variety of
 - Bahya Krimi;
 - Shonitaja Krimi;
 - Shleshmaja Krimi;
 - Purishaja Krimi

Answer: (c)

- Yuka and Pipilika are
 - Bahya Krimi;
 - Shonitaja Krimi;
 - Shleshmaja Krimi;
 - Purishaja Krimi

Answer: (a)

- Which type of Krimi is responsible for the Lakshana similar to Kushtha (skin disease)?

- Bahya Krimi;
- Shonitaja Krimi;
- Shleshmaja Krimi;
- Purishaja Krimi

Answer: (c)

- The number of Krimi explained by Sharangadhara is
 - 20;
 - 21;
 - 2;
 - 22

Answer: (b)

- Jvara, Vivarnata, Shoola, Hridroga, Bhrama, Bhakta Dvesha and Atisara are seen in

- Kushtha;
- Raktapitta;
- Vishtambha;
- Krimi

Answer: (d)

- Gudakandu is caused by

- Kaphaja Krimi;
- Raktaja Krimi;
- Purishaja Krimi;
- Bahya Krimi

Answer: (c)

- Audumbara and Romadvipa Krimi belong to
 - Raktaja Krimi;
 - Purishaja Krimi;
 - Malaja Krimi;
 - None of the above

Answer: (a)

- Sanjata Krimi Lakshana is
 - Pravahika;
 - Shotha;
 - Bhakta Dvesha and Atisara;
 - Baddha Vitkata

Answer: (c)

- Which one of the following manifested in an individual due to the presence of Krimi?

- Shvasa;
- Hikka;
- Kasa;
- Shula

Answer: (d)

- As per Madhava Nidana, Romavidhvamsa belongs to
 - Kaphaja Krimi;
 - Raktaja Krimi;
 - Bahya Krimi;
 - Purishaja Krimi

Answer: (b)

- The mean daily loss of blood due to one ancylostoma duodenale is
 - 0.15 ml;
 - 0.01 ml;
 - 0.03 ml;
 - 1.25 ml

Answer: (a)

- Which of the following is a round worm?

- Enterobius vermicularis;
- Hymenolepis nana;
- Ascaris lumbricoides;
- Tricuris trichuria

Answer: (c)

- The colour of Raktaja Krimi is
 - Like Indragopa;
 - Black;
 - Like Tamra;
 - Greenish blue

Answer: (c)

Chapter resources

- Charaka Samhita, Vimana Sthana – 7th chapter – Vyadhita Rupiya Vimana Adhyaya
- Sushruta Samhita, Uttara Tantra 54th chapter – Krimi Pratishedha Adhyaya
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